4.3 Thrust 3: From Spatial Omics Atlases to Predictive Virtual Cell Models

Overview: We aim to make existing spatial omics atlases more accessible and computationally useful. In *Task 3.1*, we will curate major spatial omics atlas datasets and develop a graphical portal to help researchers search, compare, and reuse these currently underused resources. These atlases were generated across different platforms, studies, and time points, with overlapping samples, inconsistent metadata, and heterogeneous data structures. Building on this harmonized resource, *Task 3.2* will develop an AI-based virtual cell model to predict cellular responses across diverse microenvironments. This model will support counterfactual simulation of cell behavior in both measured and unobserved tissue environments.

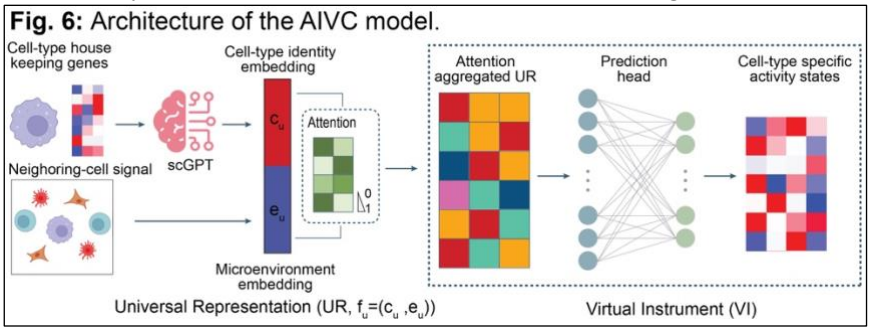


Fig. 6: Architecture of the AIVC model.

Task 3.2: We will develop an AI-based Virtual Cell (**AIVC**) model for predicting cellular activities under counterfactual microenvironments using single-cell-resolution ST data. Although ST captures cells as static snapshots, these snapshots collectively represent many cells exposed to diverse microenvironments and biological conditions. Our rationale is that, when trained at scale, static spatial transcriptomics snapshots can reveal the dynamic relationship between a cell's activity and its microenvironment. As shown in **Fig. 6**, the AIVC model has two main components: a universal representation (**UR**) that encodes both cell identity and local environment features, and a task-specific virtual instrument (**VI**) that predict cellular activity dynamics.

Step 1. Construct the UR: For each cell u, its UR f_u contains two components: cell identity embedding and microenvironmental context. The cell identity embedding c_u captures relatively stable cell-type information. To construct c_u, we first cluster cells, identify cell-type marker genes through differential expression analysis, and encode each cell's marker profile using a genomics foundation model such as scGPT[1] or UCE[2]. The microenvironment embedding e_u summarizes the the local expression environment around cell u. We will identify all neighboring cells of cell u within a predefined radius $r = 20\mu m$. For each cell type $m \in \{1, 2, \dots, M\}$, we will identify all cell u's neighboring cells that belong to that type and compute a distance-weighted average of their expression, giving greater weight to cells closer to u. This produces a cell-type-stratified microenvironment representation as $e_u = (e_{u,1}, e_{u,2}, \dots, e_{u,M})$. The final UR combines cell identity and microenvironmental context as $f_u = (c_u, e_u)$.

Step 2. Train a VI to predict cell-type specific activity states: Within each cell type, we will identify activity-state genes that distinguish functional substates among cells of the same cell-type. Because these activity programs are cell-type specific, the prediction target y_u will be the expression vector of the selected activity-state genes for the cell type of cell u. We will train a VI to take the universal representation f_u as input and predict the cell-type-specific activity-state vector $\hat{y}_u$.

As shown in **Fig. 6**, the VI begins with an attention layer that adaptively integrates the cell identity embedding and neighborhood expression features. For each cell u, the UR f_u is reweighted by feature-specific attention scores as: $h_u^i = q^i \times \tanh(f_u^i)$, $a_u^i = \exp(h_u^i)/\sum_{i=1}^{4g} \exp(h_u^i)$, where q^i where q^i is a trainable parameter for the i-th feature in the UR. The attention-weighted representation is then computed as: $\sum_{i=1}^{4g} a_u^i \times f_u^i$. Importantly, the attention scores a_u^i are interpretable. They reveal which features, including the cell's own identity markers or specific neighboring cell-type expression profiles, have the strongest influence on the predicted cellular activity state. The model will be trained using gradient descent, with a loss function containing two components, an MSE term that measures the difference between predicted and observed activity-state gene expression, and a sparsity penalty that encourages the attention layer to focus only on informative UR features by pushing irrelevant attention weights toward zero.

Step 3. Predict cell activity under counterfactual microenvironments: After training, we will use the VI to predict how a cell's activity change under alternative microenvironments. For a given cell u, we will keep its identity embedding c_u fixed and modify only its microenvironment embedding e_u. For example, we can simulate asimulated microenvironment e'_u, with the gain or loss of a neighboring cell type, changes in nearby cells' activity states, or relocation of the cell into a different neighborhood. The model will then predict the counterfactual activity state: $\Delta y_u = VI(c_u, e'_u) - VI(c_u, e_u)$, which quantifies the predicted change in cellular activity. This framework will help researchers generate testable hypotheses about spatial cell-cell interactions, prioritize microenvironmental factors that regulate cellular states, and evaluate experimental designs in silico before conducting costly wet-lab experiments.

Expected results and evaluation plan (Tasks 3.2): We will split cells into training and validation sets and match each training cell to a validation cell with similar identity but a different microenvironment. We will replace the training cell's microenvironment embedding with that of the matched validation cell and use the trained AIVC model to predict its activity state. The prediction will be compared with the matched validation cell's observed activity state to evaluate microenvironment-driven activity changes.

Potential problems and alternative solutions (Task 3.2): The AIVC model contains several tunable elements, including the spatial radius r used to define neighboring cells and the architecture of the VI-prediction network. We will systematically test these settings to optimize prediction performance.

Reference

1. Cui, H., et al., *scGPT: toward building a foundation model for single-cell multi-omics using generative AI.* Nature methods, 2024. **21**(8): p. 1470–1480.
2. Rosen, Y., et al., *Universal cell embedding provides a foundation model for cell biology.* Nature, 2026: p. 1–9.